



Effects of ambient particulate matter on fasting blood glucose: A systematic review and meta-analysis[☆]

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ABSTRACT

Studies have found that ambient particulate matter (PM) affects fasting blood glucose. However, the results are not consistent. We conducted a systematic review and meta-analysis to determine the relationship between PM with an aerodynamic diameter of 10 μm or less (PM_{10}) and PM with an aerodynamic diameter of 2.5 μm or less ($\text{PM}_{2.5}$) and fasting blood glucose. We searched PubMed, Web of Science, the Wanfang Database and the China National Knowledge Infrastructure up to April 1, 2019. A total of 24 papers were included in the review, and 17 studies with complete or convertible quantitative information were included in the meta-analysis. The studies were divided into groups by PM size fractions (PM_{10} and $\text{PM}_{2.5}$) and length of exposure. Long-term exposures were based on annual average concentrations, and short-term exposures were those lasting less than 28 days. In the long-term exposure group, fasting blood glucose increased 0.10 mmol/L (95% CI: 0.02, 0.17) per 10 $\mu\text{g}/\text{m}^3$ of increased PM_{10} and 0.23 mmol/L (95% CI: 0.01, 0.45) per 10 $\mu\text{g}/\text{m}^3$ of increased $\text{PM}_{2.5}$. In the short-term exposure group, fasting blood glucose increased 0.02 mmol/L (95% CI: -0.01, 0.04) per 10 $\mu\text{g}/\text{m}^3$ of increased PM_{10} and 0.08 mmol/L (95% CI: 0.04, 0.11) per 10 $\mu\text{g}/\text{m}^3$ of increased $\text{PM}_{2.5}$. Further prospective studies are needed to explore the relationship between ambient PM exposure and fasting blood glucose.

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1. Introduction

Diabetes is a chronic disease caused by defects in insulin secretion or function; it is associated with long-term damage, dysfunction, and failure of various organs, especially the eyes, kidneys, nerves, heart, and blood vessels (Expert Committee on the Diagnosis and Classification of Diabetes Mellitus, 1997). Approximately 451 million adults worldwide had diabetes in 2017, and this number is expected to rise to 693 million by 2045. In 2017, diabetes accounted for at least \$850 billion in health care expenditures worldwide (Cho et al., 2018). In addition to the established risk factors for diabetes, such as a family history of diabetes mellitus, age, obesity, and physical inactivity (Fletcher et al., 2002), particulate matter (PM), which has been defined as a major cause of the global burden of disease (GBD 2015 Risk Factors Collaborators,

2016), was recently linked to diabetes (Balti et al., 2014; Eze et al., 2015; Meo et al., 2015; Park and Wang, 2014; Rajagopalan and Brook, 2012; Wang et al., 2014).

Fasting glucose is not only used to define diabetes but is also an indicator for prediabetes. Individuals with impaired fasting glucose, meaning elevated fasting blood glucose levels that have not reached the level of diabetes, are considered to have prediabetes, which indicates a high risk for the future development of diabetes (American Diabetes Association, 2014). Since 2010, several population-based studies have found that ambient PM exposure has a negative influence on fasting blood glucose. However, these studies were conducted in a limited number of countries, and the results are not consistent. For example, Erqou et al. (2018) and Alderete et al. (2017) found opposite results when investigating the impact of long-term exposure to PM with an aerodynamic diameter of 2.5 μm or less ($\text{PM}_{2.5}$) on fasting glucose. Furthermore, the results were inconsistent in subgroups of the population, which will affect the priorities of future studies.

Thus, considering the importance of this topic and the varying results, we pooled evidence from relevant epidemiological studies to investigate the relationships of PM_{10} (PM with an aerodynamic

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diameter of 10 μm or less) and $\text{PM}_{2.5}$ with fasting blood glucose in this systematic review and meta-analysis.

2. Method

2.1. Data sources and searches

We searched PubMed, Web of Science, the Wanfang Database and the China National Knowledge Infrastructure for both English and Chinese articles using the search command ("particulate matter" OR "fine particulate matter" OR " PM_{10} " OR " $\text{PM}_{2.5}$ " OR "air pollution" OR "air pollutants" OR "dust") AND ("glucose" OR "blood glucose" OR "blood sugar") up to April 1, 2019, without a specific beginning date. We also examined the references of the articles that were included in the review. Full texts were reviewed for potentially relevant articles, which were selected based on titles and abstracts.

2.2. Study selection

Studies that met the following eligibility criteria were included in the systematic review: 1) original studies and 2) population-based articles describing the relationship between ambient PM concentrations and fasting glucose. For the meta-analysis, the same criteria were used, and the following additional criteria were applied: articles using the concentrations of ambient PM as the exposure variable and reporting changes in fasting blood glucose (Δ value or % change) as the outcome with complete 95% confidence interval (95% CI) information. Studies were excluded if they 1) only investigated the relationship between ambient PM concentration and diabetes without including specific blood glucose levels or 2) were review articles on this topic. The study selection process is shown in Fig. 1. We followed the PRISMA guidelines, and the PRISMA checklist is provided in the Supplementary Material.

2.3. Data extraction and quality assessment

The quality of these studies was assessed using the Newcastle-Ottawa Scale (NOS). A "star system" was developed in which a study is judged on three broad perspectives: the selection of the study groups, the comparability of the groups, and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies, respectively. A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be assigned for comparability (Wells et al., 2013). Higher scores corresponded to higher quality. The results are shown in Table 1 and Table S1.

Various measures are used to evaluate the blood glucose level, and we selected the results related to fasting blood glucose in the studies. For the articles included in the meta-analysis, we extracted one result from the main analysis instead of the stratified analysis of every article. We used the results after adjustments for confounding factors. Because of the limited number and various lags of the articles focusing on short-term exposure to ambient PM, for main results with different lags, we selected the estimated effect of the shortest lag. The data extracted from the studies included the last name of the first author, publication year, region and country of the study, study population, sample size, exposure, exposure assessment method, study design, controlled variables, effect sizes and 95% CIs. We tried to the greatest extent possible to contact authors if detailed quantitative information was not provided in the articles. Because the form that each study used to estimate the effect was different, all of the estimated effects are expressed as glucose concentrations (mmol/L) and were converted to a common exposure unit of a 10 $\mu\text{g}/\text{m}^3$ change in the concentration of PM_{10} or

$\text{PM}_{2.5}$, which allowed us to quantitatively pool estimates from different studies. Some studies used log-transformed outcomes for normality; we multiplied the percent change by the mean blood glucose level to obtain the approximate level of blood glucose change. We present the basic characteristics of the studies that were not included in the meta-analysis (Table S1).

2.4. Data synthesis and analysis

Two size fractions of PM are commonly reported in the literature (PM_{10} and $\text{PM}_{2.5}$), and two exposure lengths are described: long-term exposure and short-term exposure. "Long-term exposure" denotes that the indicator can represent a long period of exposure to ambient PM, such as an annual concentration; "short-term exposure" denotes that the indicator can represent an exposure period shorter than 28 days, which has been frequently used as a time window for medium-term exposure in several studies (Lucht et al., 2018; Peng et al., 2016). Medium-term exposure was defined as 28–91 days according to a previous study (Lucht et al., 2018), but a medium-term window was not considered in this study due to data availability. Thus, we divided the articles into four groups in our meta-analysis: 1) long-term exposure to PM_{10} , 2) short-term exposure to PM_{10} , 3) long-term exposure to $\text{PM}_{2.5}$, and 4) short-term exposure to $\text{PM}_{2.5}$. We classified these studies according to the size fractions of PM and the exposure lengths described explicitly in the text.

For the articles included in the meta-analysis, we calculated the pooled effect sizes for each group using a random-effects model; ultimately, we obtained pooled effects for 10 $\mu\text{g}/\text{m}^3$ increments of ambient PM exposure. We quantified heterogeneity using the Cochran Q (χ^2) statistic ($P < 0.10$ was considered significant) and the I^2 statistic. Heterogeneity among studies was quantified by the coefficient of inconsistency (I^2) and classified as low ($< 25\%$), moderate ($25\text{--}75\%$), or high ($\geq 75\%$). Funnel plots and Egger's test were used to evaluate the potential effect of publication bias.

To evaluate the moderating effects of age, sex and geography, we conducted a meta-regression for the long-term exposure group based on the available data. The mean age of the populations, the proportion of males in the population and the continent of the study were chosen as separate indicators. Random-effects regression with the empirical Bayesian technique was used in this process.

For the sensitivity analysis, we tested the stability of the pooled results by sequentially removing every article from each group. If the results of the meta-analysis are consistent, they show high stability and credibility. We also conducted an analysis by deleting articles that were log-transformed for stability, and the results are presented in the Supplementary Material.

All analyses were conducted using the R software (version 3.4.1) metafor package.

3. Results

3.1. Included studies

A total of 1662 articles were identified during the systemic search. A total of 1370 articles were excluded after the title or abstract was read. Among the remaining 282 articles, 24 articles were included after the full texts were read, and 17 articles were retained for the meta-analysis. Of the studies included in the meta-analysis, six examined long-term exposure to PM_{10} , four investigated short-term exposure to PM_{10} , eight assessed long-term exposure to $\text{PM}_{2.5}$, and three examined short-term exposure to $\text{PM}_{2.5}$ (Fig. 1).

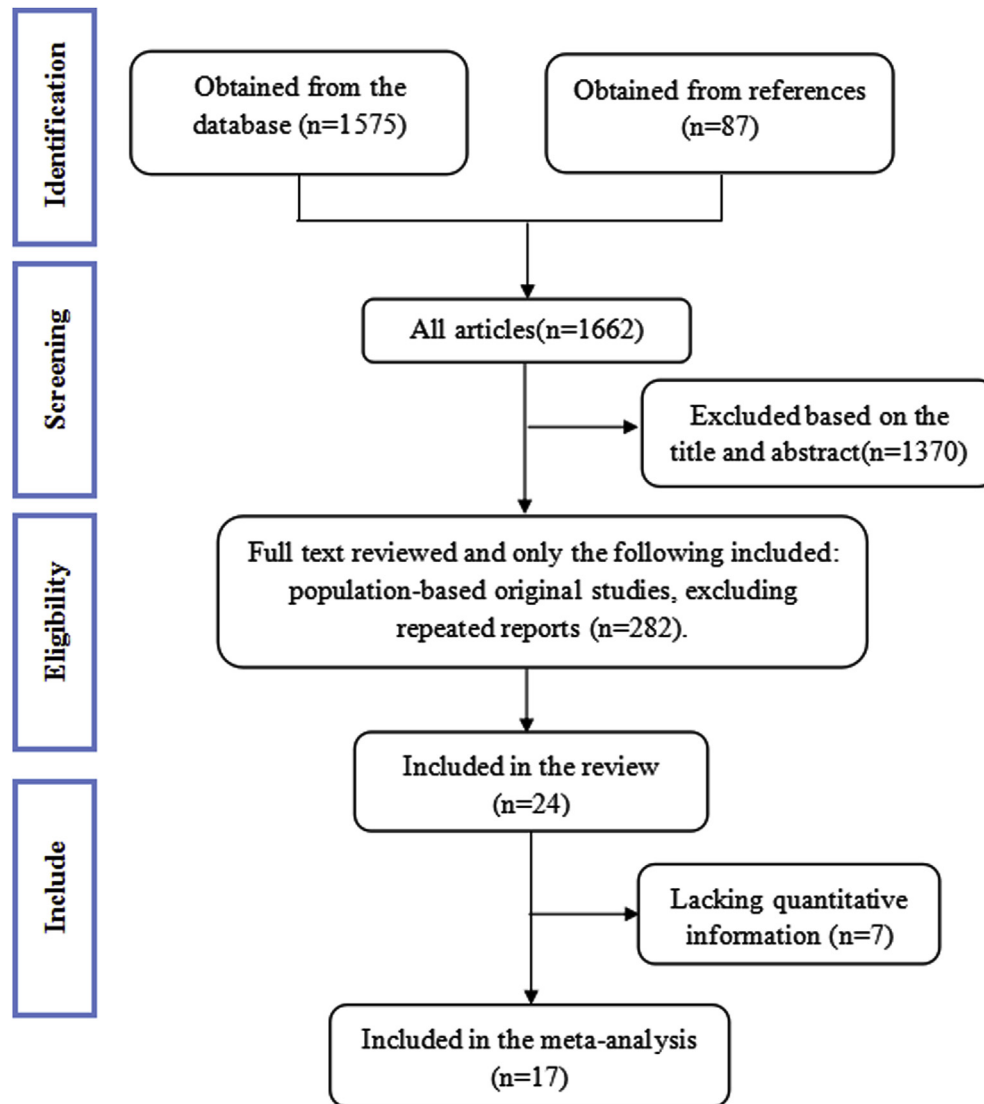


Fig. 1. Study selection process.

3.2. Characteristics of the studies included in the meta-analysis

Of all the included studies, nine had a cross-sectional design, six were longitudinal studies, one had an experimental design, and one was a repeated measures study (Table 1). Regarding the regions and countries where these studies were conducted, six studies were conducted in China, five were conducted in the United States, two were conducted in Germany, one was conducted in France, one was conducted in the Netherlands, one was conducted in Korea, and one was conducted in India (Fig. 2). The study populations examined in these reports varied from the general population to middle-aged individuals, an elderly population, pregnant women, and other groups (Table 1).

For the indicators of exposure, the PM_{10} or $PM_{2.5}$ concentration was the main research index. The exposure data were mainly collected from air quality monitors or modeling models, including land use regression (LUR) models, spatial statistical models and dispersion models. The outcome indicators mainly included fasting blood glucose and postprandial blood glucose, while some studies also used HbA1c as an outcome measure. Possible confounding factors in these studies included age, sex, BMI, diet, exercise,

smoking status, drinking status, season, date and participant socioeconomic status. For the data analyses, most studies used multivariable linear regression, mixed linear models and generalized additive models to analyze the relationships of PM_{10} and $PM_{2.5}$ with blood glucose (Table 1).

3.3. Characteristics of the studies not included in the meta-analysis

Of the seven studies that were not included in the meta-analysis, four had a cross-sectional design, two were longitudinal studies, and one had an experimental design. Three studies were conducted in the U.S., two were conducted in Israel, and two were conducted in China (Table S1). The study populations included overweight and obese youth, college students, women with gestational diabetes (GDM), the general adult population, elderly individuals, young-onset type 2 diabetes patients and populations diagnosed with chronic diseases. Five studies focused on long-term exposure, four focused on short-term exposure, and one examined medium-term exposure. Most of these articles showed a significant relationship between ambient PM exposure and elevated fasting glucose: most focused on ambient PM concentrations, whereas one

Table 1
Characteristics of the studies included in the meta-analysis.

Author	Year	Region	Population	Sample size	Design	Exposure	Exposure assessment method	Confounding factors	Results	Score
Liu et al. (2016)	June 2011 to March 2012	China	Middle-aged and elderly	11,847	Cross-sectional	Average concentrations during the study period	Spatial statistical model (10 km × 10 km)	Sociodemographic characteristics (age, sex, BMI, educational level and location of residence), behavioral variables (smoking, drinking and indoor air pollution) and ambient O ₃	An IQR increase in PM _{2.5} was associated with elevated levels of fasting glucose (0.26 [95% CI: 0.19, 0.32] mmol/L)	7
Chuang et al. (2011)	2000	Taiwan, China	Elderly	1023	Cross-sectional	1-year average	Monitoring stations	Age, sex, BMI, current smoker and drinker, visit date and yearly average temperature	Fasting glucose was significantly associated with 1 year average increases in particulate air pollution (PM _{2.5} and PM ₁₀)	6
Wolf et al. (2016)	2006–2008	Augsburg, two adjacent rural counties, Germany	General population	2944	Cross-sectional	Mean annual levels	Land use regression (LUR) models	Age, sex, smoking status, BMI, waist-to-hip ratio, month of blood collection occupational status, years of education, income, socioeconomic status, years and pack-years of smoking, physical activity, and alcohol intake	PM _{2.5} was borderline significant for glucose. For the geometric mean, a 5–95th percentile difference increase in PM _{2.5} was associated with a 1.6% increase in glucose levels in the general population	7
Cai et al. (2017)	2006–2013	Northern Netherlands	Residents aged 25–50 years	144,082	Cross-sectional	Annual concentrations	LUR models (100 m × 100 m)	Age, sex, season of blood collection, smoking status and pack-years, education, employment, alcohol consumption, daytime noise	A higher IQR for PM ₁₀ (2.4 mg/m ³) was associated with 0.6% (95% CI: 0.4–0.7%) higher fasting glucose with a remaining robust adjustment for air/noise pollution	7
Yang et al. (2018)	2009.04–2009.12	Liaoning, China	18- to 74-year-olds	15,477	Cross-sectional	3-year (2006–08) average	Monitoring stations for PM ₁₀ and a spatial statistical model for PM _{2.5}	Age, sex, BMI, education, family income, smoking, alcohol consumption, exercise, low-calorie and low-fat controlled diets, sugar-sweetened soft drink consumption, family history of diabetes, and district (or community)	PM ₁ , PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , and O ₃ were associated with the concentrations of fasting glucose (0.04–0.09 mmol/L) and 2-h glucose (0.10–0.19 mmol/L). The associations between air pollutants and glucose and insulin homoeostasis markers appeared to be generally stronger for PM ₁₀ and NO ₂	8
Erqou et al. (2018)	2001–2014	The greater Pittsburgh metropolitan area, PA, USA	45- to 75-year-olds	1717	Cohort study	1-year average	LUR models (300 m × 300 m)	Age, sex, smoking status, race, income and education status	Linear increase in the mean blood glucose, which remained statistically significant after adjustments for age, sex, smoking status, and race	7
Khafaie et al. (2017)	2005–2007	Pune, India	Young-onset T2DM patients	1213	Cross-sectional	1-year average	Atmospheric dispersion model (2 km × 2 km)	Age, sex, years of education, season of enrollment, BMI, WHR, duration of diabetes, diet, alcohol consumption and smoking	1 SD increase in PM ₁₀ (43.83 µg/m ³) was significantly associated with 2.25 and 0.38 mmol/L increases in the arithmetic means of HbA1c and 2hPG, respectively. An association between PM ₁₀ and FPG was not found	6
Riant et al. (2018)	2011–2013	Northern France	Adults (aged 40–65 years)	2895	Cross-sectional	Annual average	Incorporation based on an atmospheric dispersion	Sex, age, urban area, the period of blood sample collection, BMI, educational level,	Associations between FBG and air pollution did not achieve statistical significance	6

Table 1 (continued)

Author	Year	Region	Population	Sample size	Design	Exposure	Exposure assessment method	Confounding factors	Results	Score
Alderete et al. (2017)	2001–2012	California, USA	Overweight and obese Latino children	314	Cohort study	Yearly average over the follow-up period	model (25 m × 25 m) Spatially interpolated from a monitoring station (50 km)	smoking status, and physical activity Tanner stage, body fat percent, season, variations in prior year exposure at each follow-up visit, study entry year	Fasting glucose levels were not associated with PM _{2.5}	7
Lu et al. (2017)	March 2006 to December 2014	Southwestern Taiwan, China	Pregnant women	3589	Retrospective study	Trimester-specific exposure; moving averages from 1 to 12 months before 100-g OGTT administration	Fixed-site monitoring station	Individual-specific effects (nulliparous status, age, BMI, season, and year) and the moving averages of temperature and relative humidity in the corresponding study period	In the single-pollutant model, significant associations were found between PM _{2.5} and fasting OGTT in all periods. The two-pollutant model yielded similar results; the preadministration 1–12 month moving averages of the IQR increases in PM _{2.5} levels were significantly associated with fasting OGTT (1.32–5.87 mg/dL)	6
Li et al. (2018)	1998–2001; 2005–2008; 2002–2005; 2008–2011	Northeastern USA	Adults	5958	Cohort study	2003 annual average and 1- to 7 day moving averages	Hybrid spatial-temporal model (1 km × 1 km)	Age, sex, BMI; smoking status, pack years, alcohol intake, educational attainment, physical activity, usual occupation, census tract-level median household income, the median value of owner-occupied housing units, population density, date of examination visit, seasonality, exam identifier, missing indicators	Participants who lived 64 m (25th percentile) from a major roadway had 0.28% (95% CI: 0.05%, 0.51%) higher fasting plasma glucose than participants who lived 413 m (75th percentile) away, and the association appeared to be driven by participants who lived within 50 m of a major roadway.	7
Lucht et al. (2018)	2000–2003; 2006–2008	Ruhr area, Germany	45- to 75-year-olds	7108	Cohort study	Residential mean exposure for the 28- and 91 day and 1, 2, 3, 7, 14, 45, 60, 75, 105, 120, and 182 day exposure windows	European Air Pollution Dispersion (EURAD) chemistry transport model (1 km × 1 km)	Temperature, humidity, examination, age, season, nutrition index, smoking status, BMI, time since last meal	Positive associations were found between PM _{2.5} and blood glucose levels. PM _{2.5} and PM ₁₀ were positively associated with HbA1c. The mean exposure levels during longer exposure windows (75–105 days) were most strongly associated with HbA1c, whereas 7–45 day exposure windows were most strongly associated with blood glucose	7
Chen et al. (2016)	2006–2008	Tangshan, China	General population	27,685	Cohort study	Daily concentrations (lag 0,0–1,0–2,0–3)	Monitoring station	Age, sex, BMI, current, drinking status, smoking status, annual family income, level of education, BP, history of diabetes and treatment, exercise activity, marital status, work type, seasonality, hospital, weather information	For exposure to 4 day average concentrations, a 100 µg/m ³ increase in PM ₁₀ was associated with a 0.11 mmol/L (95% CI: 0.07–0.15) increase in FBG. The effects of air pollutants on FBG were stronger in female, elderly, and overweight people than in male, young and underweight people	6
Chuang et al. (2010)	2002	Taiwan, China	General population	26,685	Cross-sectional	Daily concentrations	Monitoring station	Age, sex, BMI, current drinking status, current smoking status, annual family income, season,	Elevation in 1-, 3-, and 5-day averaged PM ₁₀ per IQR was not significantly associated	6

(continued on next page)

Table 1 (continued)

Author	Year	Region	Population	Sample size	Design	Exposure	Exposure assessment method	Confounding factors	Results	Score
Kim and Hong (2012)	2008–2010	Seoul, South Korea	Elderly	560	Longitudinal panel study (repeated measurement times varied, weighting following observations)	Daily mean concentrations (lag 0.0–1.0–2.0–3.0–4.0–5.0–6.0–7.0–8.0–9.0–10)	Monitoring station	geographic areas, visit date and daily temperature	with fasting glucose [–0.44 (–1.49 to 0.60), 0.25 (–0.76 to 1.27), 0.50 (–0.38 to 1.38) mg/dL] Significant associations were observed for IQR increases in PM ₁₀ , with the strongest associations observed on lag day 4 for 0.11 mmol/L; 95% CI: 0.05, 0.17. Associations were more apparent among participants with a history of DM	7
Brook et al. (2013)	Summer months of 2009 and 2010	Michigan, USA	Healthy nonsmoking subjects	25	Experimental study	Daily 4- to 5-h ambient air pollution exposure	Monitoring station	Age	A 10 µg/m ³ increase in subacute PM _{2.5} exposure was associated with increased glucose (5.4 mg/dL, 95% CI: 0.5 to 10.3) after adjustment for age	6
Peng et al. (2016)	2000–2011	Greater Boston area, USA	Elderly	551	Cohort study	Daily concentrations	Hybrid land-use regression satellite-based model (10 km × 10 km)	Age, BMI, race, regular patterns of physical activity, smoking status, cumulative pack-years of smoking, alcohol consumption, education level, statin use, temperature, seasonality	Interquartile increases in 1-, 7-, and 28-day PM _{2.5} exposure were associated with 0.57 mg/dL (95% CI: 0.02, 1.11, P = 0.04), 1.02 mg/dL (95% CI: 0.41, 1.63, P = 0.001), and 0.89 mg/dL (95% CI: 0.32, 1.47, P = 0.003) higher FBG, respectively	7

study focused on traffic-related concentrations, which also showed a positive relationship with fasting glucose. We were unable to conduct a meta-analysis of these studies because quantitative or accurate information related to the results was lacking (for example, one article was lacking 95% CIs and only reported P-values), or because the reported information was not sufficient to convert the percent change to the level of change in blood glucose. When indicators other than PM₁₀ or PM_{2.5} concentrations, such as the distance from a major road, were used to evaluate the relationship, we excluded the article from the meta-analysis.

3.4. Primary meta-analysis

For long-term ambient PM exposure, a 10-µg/m³ PM₁₀ increase was associated with a 0.10-mmol/L (95% CI: 0.02, 0.17) increase in fasting glucose (Fig. 3 A), and a 10-µg/m³ PM_{2.5} increase was associated with a 0.23-mmol/L (95% CI: 0.01, 0.45) increase in fasting glucose (Fig. 3 B). For short-term exposure to ambient PM, a 10-µg/m³ PM₁₀ increase was associated with a 0.02-mmol/L (95% CI: –0.01, 0.04) increase in fasting glucose (Fig. 3 C), and a 10-µg/m³ PM_{2.5} increase was associated with a 0.08-mmol/L (95% CI: 0.04, 0.11) increase in fasting glucose (Fig. 3 D).

The percentages of total variation across these studies caused by heterogeneity were high: 95.35% for long-term exposure to PM₁₀ (P < 0.01), 99.41% for long-term exposure to PM_{2.5} (P < 0.01), 85.64% for short-term exposure to PM₁₀ (P = 0.01), and 0.25% for short-term exposure to PM_{2.5} (P = 0.19). These results reflect the inconsistent findings among the studies. Egger's test found no

evidence of a publication bias for PM₁₀ or PM_{2.5} for either exposure period (P = 0.18 and 0.13 for long-term exposure to PM₁₀ and PM_{2.5}, P = 0.58 and 0.30 for short-term exposure to PM₁₀ and PM_{2.5}, respectively) (Figure S1).

3.5. Meta-regression

The meta-regression of age, sex and geography on the effects of elevated fasting glucose levels are shown in Table 2. The effects of a 1-year increase in mean age on the risk of elevated fasting glucose were significant in the case of long-term exposure to PM₁₀ (0.0148, 95% CI: 0.0071, 0.0225), and a 1% increase in the proportion of males in the population was also associated with the effects of long-term exposure to PM₁₀ (–0.0262, 95% CI: –0.0494, –0.0029). These two factors have no significant effects in the long-term exposure PM_{2.5} group. No significant effects of geography were found in the long-term exposure group. The results showed that the heterogeneity of the meta-analysis may be attributable to the age and sex distribution of the study population and other unknown factors.

3.6. Sensitivity analysis

We extracted the estimated effect up to 18 times sequentially, and in most cases, we did not find an obvious change in heterogeneity (Tables S2–S4). The short-term PM_{2.5} exposure group was limited in number; thus, we could not perform a sensitivity analysis. We also performed the analysis after the deletion of the log-transformed results (Table S5), and the findings were consistent

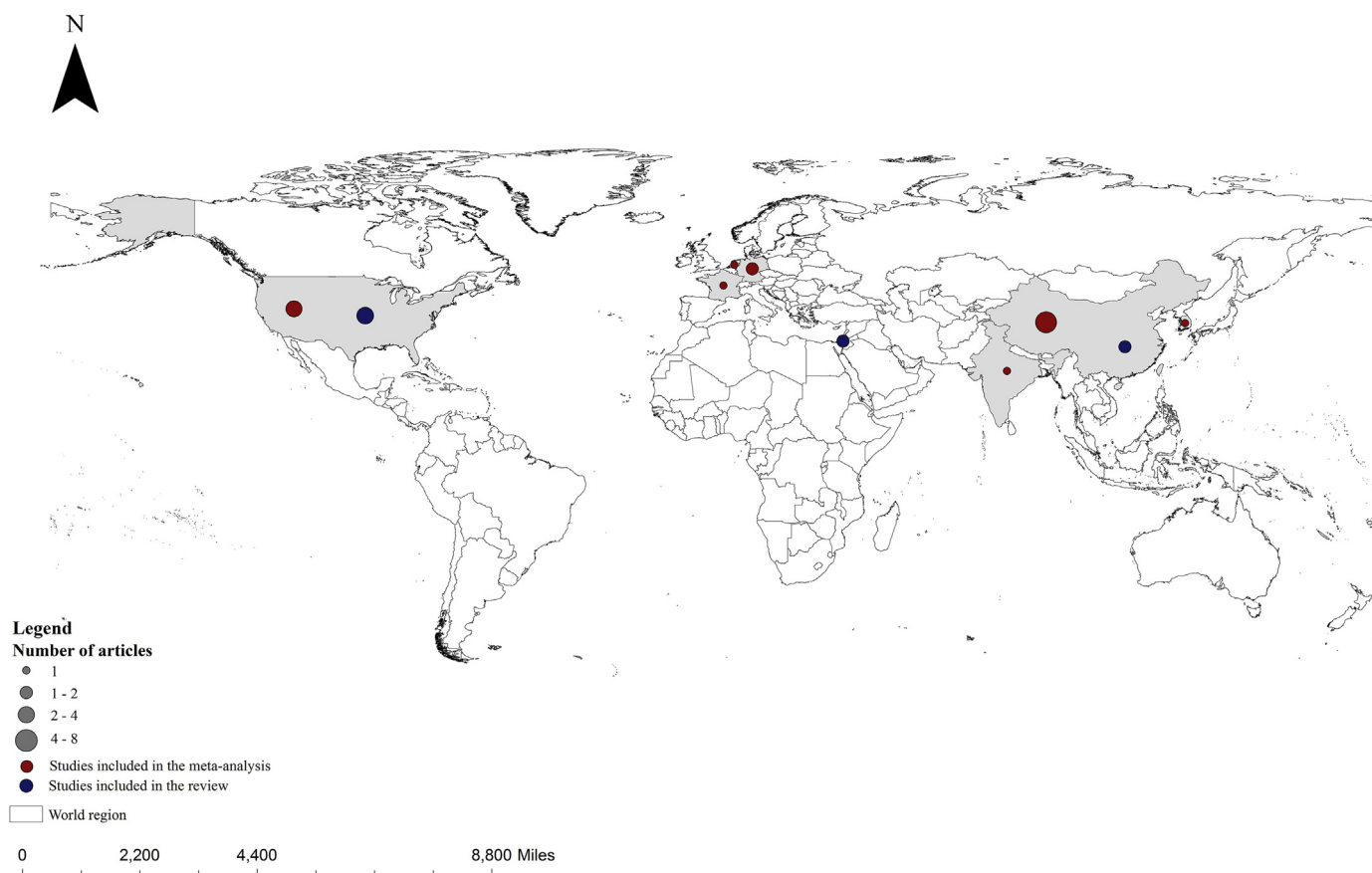


Fig. 2. Countries of the studies included in the systematic review and meta-analysis.

with the main results.

4. Discussion

To the best of our knowledge, this is the first systematic review and meta-analysis focusing on the relationships of PM_{10} and $PM_{2.5}$ with fasting blood glucose. We found that elevated fasting blood glucose was statistically associated with long-term exposure to both PM_{10} and $PM_{2.5}$ and short-term exposure to $PM_{2.5}$. We obtained conclusions regarding exposure assessments, exposure time windows, susceptible populations and possible mechanisms.

We focused on associations between fasting glucose and different forms of PM. Comparisons of the effect size of PM_{10} and $PM_{2.5}$ were complex, and we were unable to draw definitive conclusions in our meta-analysis. Smaller particles may provide a proportionally larger surface area, resulting in potentially increased concentrations of adsorbed or concentrated toxic air pollutants per unit mass (Qiu et al., 2012). Studies have focused on the negative effects of PM_1 (Yang et al., 2018), which has a smaller particle size. The current research studies are from various parts of the world, and the air pollution levels and population characteristics varied significantly. If a sufficient number of studies in a specific area become available for a future meta-analysis, the results may provide more accurate data to explain the differences in the effects of PM_{10} and $PM_{2.5}$ on fasting blood glucose.

Most studies assessed ambient particulate matter concentrations collected from monitoring stations (Alderete et al., 2017; Brook et al., 2013; Chen et al., 2016a; 2016b; Chuang et al., 2010; Chuang et al., 2011; Kim and Hong, 2012; Lu et al., 2017; Sade et al., 2015; Toledo-Corral et al., 2018; Yang et al., 2018). For studies

focusing on fasting blood glucose, which is an individual-level indicator, exposure data at the area level may lead to misclassification, which may result in a decrease in the magnitude of the associations (Goldman et al., 2011). In addition to determining exposure at a specific address to increase the accuracy of the exposure assessment, several studies used LUR models (Erqou et al., 2018; Wolf et al., 2016; Cai et al., 2017), chemical transport models (Lucht et al., 2018) and atmospheric dispersion models (Khafaie et al., 2017; Riant et al., 2018) to assess exposure. Several studies merged grid cells with high spatial and temporal resolution based on satellite data (Li et al., 2018; Liu et al., 2016; Peng et al., 2016; Sade et al., 2016; Wallwork et al., 2017; Yang et al., 2018). Li et al. (2017) and Jiang et al. (2016) assessed participants' individual exposure levels. Currently, machine learning algorithms, such as random forest models (Chen et al., 2018; Liu et al., 2018), are very popular due to their high performance for estimating air pollutant concentrations, especially in studies focusing on long time periods and large research areas. Researchers should select an appropriate exposure assessment method after considering their study's purpose. Different exposure assessment methods impact the reported outcomes, which have been taken into consideration in some articles (Khreis and Nieuwenhuijsen, 2017; Giannini et al., 2017). Furthermore, various periods of exposure, lags and the lack of a time-activity mode will increase exposure bias. Given the rapid increase in articles on this theme, it may become possible to assess the effects of the exposure assessment method and exposure periods and lags in the future.

For the studies included in the short-term effect group, the exposure period that we defined was shorter than 28 days. The exposure period varied from 0 to 28 days, and the trend was not

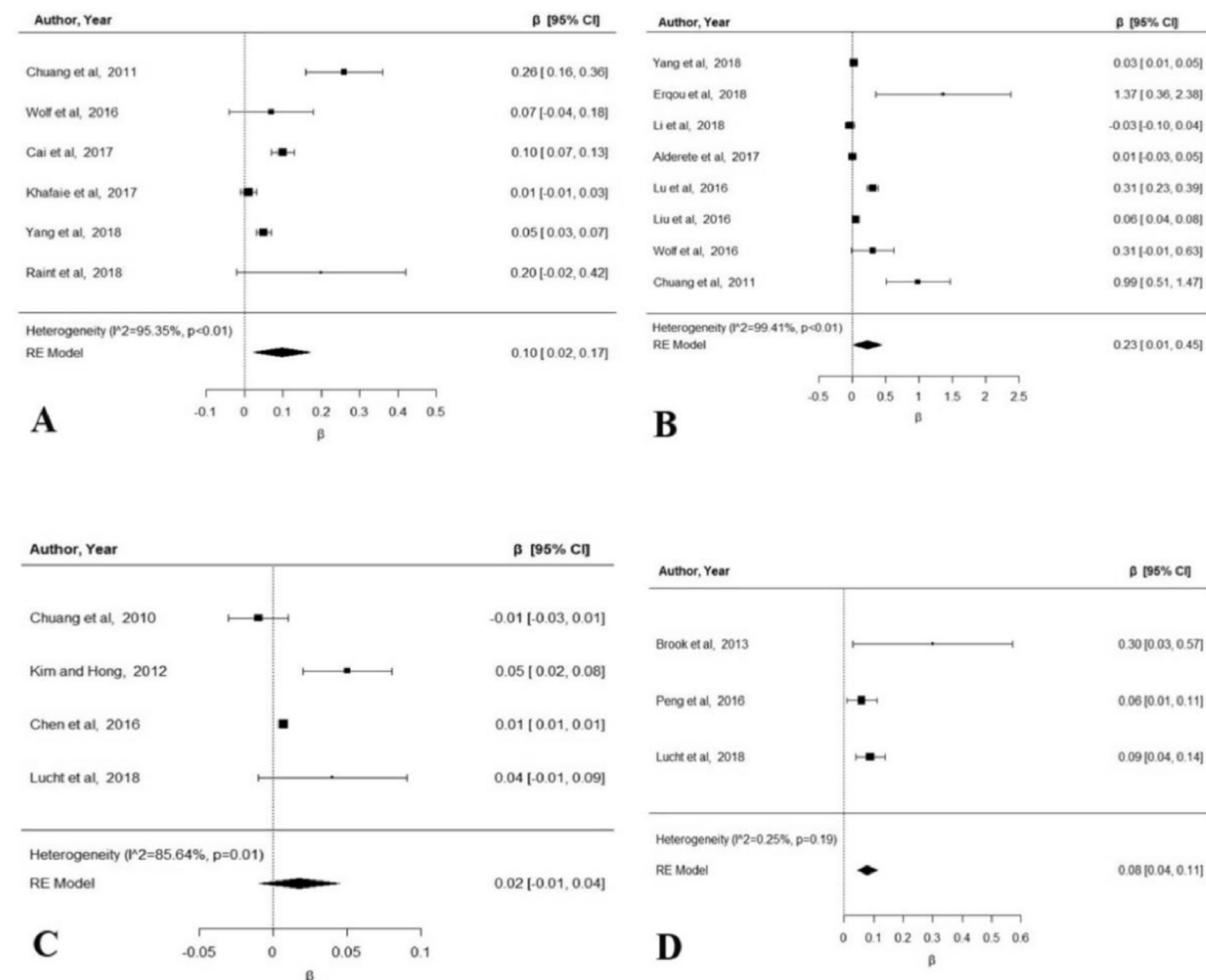


Fig. 3. Results of the meta-analysis*. Results of (A) long-term exposure to PM₁₀, (B) long-term exposure to PM_{2.5}, (C) short-term exposure to PM₁₀, (D) short-term exposure to PM_{2.5}. * Unit in 10 $\mu\text{g}/\text{m}^3$.

Table 2
The results of meta-regression in the long-term exposure group.

Group	Age		Male proportion		Geography	
	β	95% CI	β	95% CI	β	95% CI
Long term exposure to PM ₁₀	0.0148	0.0071, 0.0225	-0.0262	-0.0494,-0.0029	0.2352	-0.0302, 0.5007
Long term exposure to PM _{2.5}	0.0137	-0.0067, 0.0341	-0.0070	-0.0247, 0.0108	0.1323	-0.4230, 0.6876

consistent. [Chen et al. \(2016a\)](#) found the highest estimated effect in lag0-1 and lag0-2, [Kim and Hong \(2012\)](#) and [Peng et al. \(2016\)](#) found the highest estimated effect in lag4 and lag 0-7. In contrast, [Lucht et al. \(2018\)](#) studied longer exposure durations and found the strongest effects with 28-, 45-, and 60-d exposure windows. Due to the limited number of studies, we could not draw conclusions about potential trends in the exposure window. For the long-term exposure group, most studies used the annual average concentration as an indicator, while some studies used the average concentration within a specific period ([Liu et al., 2016](#); [Lu et al., 2017](#)). Lag days strongly influenced the results and may reveal potential

exposure windows for studying the relationship between ambient PM exposure and fasting glucose; thus, more studies, especially studies focusing on acute effects, are required to address this issue in the future.

We found that a higher average age level had effects on increased fasting glucose with long-term exposure to PM₁₀, possibly because elderly individuals are more prone to IR and loss of islet β cell compensatory function; additionally, lifestyle and medication factors may also impact the relationship ([Liu, 2005](#)). A decreased proportion of males in the study population was also associated with increased fasting glucose, possibly due to

differences in airway structures and social status in females (Chen et al. (2016a)). The conclusion that elderly people and females experienced stronger effects conflicted with the results of some previous studies (Chen et al., 2016a; Liu et al., 2016; Yang et al., 2018). More studies focusing on differences between sexes and age groups are needed to clarify the relationship between ambient PM exposure and fasting blood glucose. Furthermore, overweight or obese individuals (Chen et al., 2016a) and individuals with low education levels (Liu et al., 2016) showed increased sensitivity to PM exposure in some studies. Some researchers found that the fasting glucose level of the diabetic population is sensitive to PM₁₀ or PM_{2.5} (Kim and Hong, 2012; Wolf et al., 2016; Li et al., 2018), while most studies excluded patients with diabetes or metabolic syndrome or individuals using hyperglycemic drugs. Researchers should consider sensitive populations in the future.

In a review of these articles, we found several possible mechanisms for the relationship between ambient PM exposure and fasting blood glucose. For acute exposure to PM, Brook et al. (2013) found that air pollution exposure directly affected IR. Li et al. (2017) proposed possible mechanisms, such as an increase in glucocorticoids caused by PM exposure leading to increased blood glucose levels. Air pollution also leads to intense oxidative stress and adipose tissue inflammation (Anderson et al., 2012; Fleisch et al., 2014). With chronic exposure, changes in endothelial function reduce insulin sensitivity and affect peripheral glucose uptake (Sun et al., 2005). Previous studies found that exposure to PM_{2.5} leads to activation of the endoplasmic reticulum stress pathway, increased liver inflammation, and abnormal insulin receptor substrate phosphorylation (Rui et al., 2016; Zheng et al., 2013). Sympathetic nervous system activation caused by exact particle exposure (Balasubramanian et al., 2013) and hypothalamic-pituitary-adrenal axis response excitement (Thomson et al., 2013) also play important roles in the mechanism underlying the relationship between PM_{2.5} and IR.

In addition to fasting glucose, HbA1c and two-hour plasma glucose levels were used as indicators in some of the included articles. The 2-h postprandial glucose (PPG) level is an important parameter for increasing the diabetes detection rate in participants with high fasting blood glucose levels (American Diabetes Association, 2014). HbA1c reflects blood glucose concentrations for 1–2 months prior to blood testing (Krishnamurti and Steffes, 2001); it has shown high sensitivity and specificity in some studies (Little et al., 1988; Rohlfing et al., 2000) and is widely used as an indicator of glycemic control in diabetic patients. Lucht et al. (2018) found that longer exposure windows were most strongly associated with HbA1c, whereas shorter windows were most strongly associated with blood glucose. Riant et al. (2018) also suggested that serum glucose may be less accurate than HbA1c as an indicator of the association between long-term exposure to air pollution and blood glucose levels, indicating possible differences between these indicators with different exposure windows. Due to the limited number of studies, we could not explore the potential relationship between ambient PM exposure and these two indicators, which are especially important for selected populations, such as those who are pregnant, diabetic, or prediabetic. More studies are needed in the future.

There are some limitations to our review. One limitation of our study is the limited number of included articles, which resulted in high heterogeneity across the groups. Biases may have been present because of inconsistent exposure assessment methods and differences in exposure lags and windows, populations, and study areas among the studies included in the meta-analysis. In addition, we could not conduct meta-regression analyses for all groups to identify the origin of the heterogeneity across studies. More prospective and standardized studies are needed in this field in the

future.

Considering the necessity and the limitations of the current researches, many more longitudinal studies that can confirm the causal relationship between ambient PM exposure and glucose levels should be carried out in the future. By using statistical methods such as machine learning or monitoring pollutants at the personal exposure level, an exposure assessment method with strong comparability and accuracy can be established to enhance our understanding of the relationship between PM exposure and blood glucose levels. Further analyses of glucose parameters that consider study aims, populations and exposure durations are also needed. More studies focusing on varied populations should be conducted in the future to verify whether specific populations are particularly sensitive to PM exposure effects.

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Author contributions

T.L. contributed to the conception and design of the study. R.M. and Y.Z. collected the data, performed the statistical analysis and drafted the article. All authors (R.M., Y.Z., Z.S., D.X. and T.L.) contributed to interpreting the results and critically revising the draft.

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Declaration of competing interest

The authors declare that they have no actual or potential competing financial interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2019.113589>.

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